

Climate Hazards: Outline

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Questions for co-authors...

1. Only max temperature matters to heat extremes – unlike hardiness where plants gain/lose tolerance, this doesn't really happen for heat extremes, right?
2. What extremes to think about? Heat, cold, drought ... ?
3. Is there any literature on heat damage as limiting seasons? Probably no; always drought.
4. Any ideas on figures?

1 Flailing around in November 2025

For now, I will confine myself here to this section in this document ...

Questions

- What are the impacts of increasing temperatures versus increasing variance in temperatures? And when is the interaction important (versus they can simply be predicted from knowing the effects of each one alone)?
- Do historical drivers extend into the future projections?

Thinking on setup ... not sure this is helpful...

- Things that could shift in general
 - Changes in frost risk
- Things that could shift with warming
 - Accelerated development (leaves, fruit), leading to fruit maturation happening more often
 - Delayed development (leaves only?) due to too much heat
- Things that could shift with SD
 - More frost events?

Methods

- We used historical climate data (YEARS, dataset) and applied shifts to the historical data to simulate increases in means and shifts in variance.
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- Drought survival should not matter as we do not change the water in the soil (WHC) or the precip.

Maybe we also want something like ... 2023Jun_MontpellierVisitnotes.pdf?

Results

- Historical trends
 - This is explained in Section 1 (incl. Fig 1 caption) in climhazardsresults.pdf ... we have no better plots so far, but you can see also:
`analyses/graphs/phenofit/historical/allsp_maindriverxy.pdf`
- Effects of increasing mean temperatures (summarized from results file, see above)
- Fagus
 - Fagus depends on latitude: at low latitudes there is reduced CarbonSurvival (not enough cold means late dormancy) and thus FruitMaturationDate gets later. While at higher latitudes there is an increase in the FruitIndex as FruitMaturation is higher.
 - Quercus – not much effect, FruitMaturation gets a little later at low latitudes due to ... ??
 - Pinus is pretty similar across latitudes, it just gets hammered by low CarbonSurvival due to low chilling and thus later dormancy break date
- Effects of SD ... (copied from results file, see above)
 - *Fagus* is determined mainly by a combo of damage to leaves and flowers, which increases with increasing variance.
 - *Pinus* survival dominates at low latitudes (no carbon survival and variance does not change this), but at highest latitude variance also leads to later leafout (low chill, later dormancy break) and thus lower leafindex (note that Pinus can sustain VERY low temperatures and is unlikely to have frost damage).
 - *Quercus* is not changing much but does see some frost losses at higher variance.
- Effects of increasing mean temperatures and shifts in variance (summarized from results file, see above)
 - Fagus depends a little on latitude: at site 41 N low variance is better except at high warming where there are issues with late breaking, at 53 N: Increased variance is constantly worse, but increased warming leads to less frost. Fruit maturation stalls (even though flowering advances) though it appears to have no or low impact on fitness, but we did look closely at it – see Fig 21 in the results file.

- *Pinus*: Variance and warming compound to lead to greater negative effects. We made some endodormancy curves, see Fig 19 of results file.
- *Quercus*: Not much happens, esp. at lower latitudes. At 53 N higher variance, lower warming leads to some frost but not much.

2 Outline

1. Introduction: mean versus variance in climate change biology
 - (a) Increasing interest in impacts of extremes
 - (b) Lots on thermal curves and variance, including lots on Jensen's inequality
 - (c) Lots of interest in shifts in variance from theoretical perspective, but often not linked to empirical reality
 - (d) Most find increasing variance will reduce fitness (e.g., Vasseur), but how accurate is this?
 - (e) As variance is a fundamental part of life history theory and physiology, it is likely more complicated
 - (f) Here we ...
2. Why extremes matter
 - (a) Overview of basics
 - i. Lots on mean versus variance theoretically in life history theory (colors of noise etc.)
 - ii. And in physiology: mean development, but also how extremes limit distributions
 - iii. Molecular studies back up this complexity: variability versus mean temperatures underlie pathways to some events
 - iv. But climate change smears across these two separate approaches as we need to predict outcomes across the full life stages of individuals within a population and across species.
 - (b) Fundamental trade-off in life history that climate change is (most?) rapidly altering: growing season length (mean) versus risk (variability often)
 - i. Frost risk
 - ii. Drought risk
 - (c) What extremes? (could move to box 2?)
 - i. Heat: just the max (right?)
 - ii. Cold
 - A. Max (hardiness at max period)
 - B. Transition periods
 - iii. Drought
 - iv. Others?
 - (d) Why biological hazards should be easier to predict than climatic hazards (transition to next section with this? Or find it new home?)
 - i. Extremes in climate science are defined statistically usually – they are rare by definition which makes them tricky
 - ii. But we're not talking about that, we're talking about climate hazards, which depend on biological limits, often thermal limits
3. When extremes matter (need Fig)

- (a) Variance shifts in certain biological periods matter a lot more than others: Likely windows for hazards ...
 - i. Transitional climatic periods (spring/fall) and hottest summer months
 - ii. But also depend a lot on phenology
- 4. PHENOFIT case study
 - (a) Do means versus variability shifts alone lead to increases or decreases in fitness? Or is it messier?
 - (b) Additive effects...
- 5. From models to forecasting (*this section needs work*)
 - (a) Need more fitness data
 - (b) Need more data on events and their impacts (crops?)
 - (c) Need better molecular studies of mean versus variability
 - (d) Need to bridge the observational/experimental gap for heat extremes (especially)

Boxes?

- 1. Box 1: Why we're bad at predicting extremes
 - (a) We may often have poor intuition about what is shifting ... How much are means versus variability shifting?
 - i. Info from the literature
 - A. Basics from IPCC?
 - B. Refs from Ben?
 - C. Extremes versus variability
 - ii. Lots of variance patterns in temperature are narrowing:
 - A. Daily temperature ranges
 - B. Elevation, latitude
 - (b) Case study in Europe
 - i. Variability across space (sites)
 - ii. Shifts since 1950 by month
 - iii. Change over time versus sites: Variability across space decreasing?
 - iv. Projections
- 2. Box 2: High versus low temperatures

3 Reference notes

? covers Jensen's Inequality; mainly just that you can get the wrong mean if you don't take into account variance; and you can predict useful stuff if you know the variance frequency and some stuff.

? is again Jensen's Inequality but this time with climate change—looks at mean vs. skew in variance studies of thermal thresholds (I think); focuses on increase in variance.

? Show that cold temperature thresholds evolve more than high temperature (so this goes with limited acclimation also for high temperatures).

Should review: Buckley & Kingsolver AREES

4 Manuscript text

Title: Seasonal pressure points of climate change

Abstract

Climate change is reshaping growing seasons globally with major impacts on natural and agricultural ecosystems. Yet we are uncertain exactly how, where, and when impacts will be most pronounced. We show how fundamental life history theory and physiology can help identify the pressure points of climate change—seasonal periods when shifts in climate interact with development to lower growth, reproduction or survival. Using an integrated model of the full annual cycle of plant growth, reproduction and survival (PHENOFIT), we will compare the impacts of future warming versus shifts in frost events on the fitness of three tree species (*Fagus sylvatica*, *Pinus sylvestris*, *Quercus robur*). This framework will help identify the challenges and opportunities in adapting to climate change across European forests.